Aman Bansal

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Academic Qualifications

Bachelor of Technology (Honours), IIT Bombay, Mumbai 2016 - Ongoing

- Major : *Computer Science and Engineering* Current CPI (after 6 semester) : ***9.64/10***
- Bachelor's Thesis (ongoing): Exploring *Boolean Functional Synthesis* to get a poly-time algorithm for factorisation
- **Advanced Courses** : Cryptography & Network Security, Principles of Concurrent and Parallel Programming (including Distributed Systems), Databases, Operating Systems, Blockchain Technology, Numerical Analysis
- **Machine Learning & AI** : AI & ML (Theory + Lab), Advanced Machine Learning (Graphical Models, Deep Learning, Reinforcement Learning), Data Analysis and Interpretation

Scholastic Achievements

- Achieved **All India Rank 1** in JEE (Advanced) out of 150,000 shortlisted candidates 2016
- Secured **All India Rank 71** in JEE Mains out of 1.2 million candidates 2016
- Amongst the top 2 in the department to receive the **Institute Academic Award** 2019
- Overall ranked **5th** in the department out of total 120 students 2019
- Stood at 17th position among 225 teams in the **ACM-ICPC** Amritapuri regionals 2017
- Awarded AP grade in Compilers Lab and Automata Theory (awarded to **top 1%** of the class) 2019

Olympiads And Scholarship

- Awarded Scholarship under the **Charpak Research Internship Program** by the Embassy of France in India to undertake research internship in France (among the 13 students selected nationally) 2018
- Received **Gold Medal** and Certificate of Merit for being in the national **top 38** candidates at INChO 2016
- Awarded Certificate of Merit for being among the **national top 1%** in NSEP, NSEC and NSEA 2015
- Recipient of the prestigious **KVPY Fellowship** by Dept. of Science and Technology, Govt. of India 2015

Internships and Research Projects

Rubrik Inc, Bangalore | *Software Engineering Intern* May 2019 - Jul 2019

- Developed an **end-to-end** feature for implementing a user-configurable behaviour of on-demand backups
- Used **Cassandra** for stable storage and a Data Access Object pattern for interacting with it
- Implemented the complete new behaviour of the backups for every supported object without compromising efficiency
- Created optimized **RESTful APIs** and exposed them to the UI to integrate them with the UI/UX

INRIA Nancy, France | *Research Intern* | *Guide: Prof. Steve Kremer* May 2018 - July 2018

- Studied about automated, symbolic verification techniques and manipulating behavioural equivalences
- Designed a simulator for interactively displaying attacks on a protocol that violate a **behavioural equivalence**
- Implemented the operational semantics of **applied pi calculus** and trace equivalence in javascript

Extending Foundations of Differential Privacy | *Guide: Prof. Manoj Prabhakaran* Jan 2019 - Ongoing
Submitted in Eurocrypt '20

- Proposed two new concepts, **Robust Privacy** and **Flexible Accuracy** along with their composition theorems, for extending the notion of differential privacy to accommodate highly sensitive functions like *maximum*
- Invented new mechanisms with optimal parameters and improved upon the **state-of-the-art** for such functions

Liveness Based Garbage Collection | *Guide: Prof. Amitabha Sanyal* Dec 2018 - Ongoing
Currently being written to be submitted in ISMM '20

- Designed a technique for **static context-sensitive** liveness analysis by exploiting recurring liveness patterns
- Improved upon the standard algorithms in garbage reclamation and collection time by an **order of magnitude**
- Currently integrating the prototype in a *Racket* compiler to observe performance gains on real-world programs

Key Projects

Learning Latent Representation of Sound | *Advanced Machine Learning* Spring 2019

- Used instantaneous frequency spectograms (**ICLR'19**) as audio representations for developing auto-encoders
- Trained **Autoencoders** and **Beta-VAE** on them for modelling latent space of samples in audioset dataset

Autonomous Teaching Assistant | *Institute Technical Summer Project* Summer 2017

- Created a platform to frame questions from text corpus using statistical Natural Language Processing (**NLP**)
- *Stanford Parser* was used for POS-tagging and Natural Language Toolkit (**NLTK**) library for parse-tree operations
- Introduced *Named Entity Recognition*, *Pronoun Resolution* and *Synonym/Antonym Detection* for better results

Parallelised Delaunay Triangulation | *Parallel Programming* Ongoing

- Speeding up the execution of the FFT algorithm by parallelising it on a GPU using CUDA and MPI toolkit in C++

Byzantine PAXOS Consensus Protocol | *Distributed Systems* Ongoing

- Implementing the PAXOS consensus protocol in a distributed system with Byzantine faults tolerance abilities

Secure Online Exam Portal | *Database and Information Systems* Autumn 2018

- Developed a platform for taking online exams using *Servlets* and *JSP* with *Tomcat* on the server side
- Used the *JDBC API* for connecting and interacting with the **PostgreSQL** RDBMS on backend
- Safeguarded the web-pages against *SQL Injection* and *Cross-site scripting* (**XSS**) attacks

D-RAM Addressing Attack on Intel i7 | *Computer Architecture* Autumn 2018

- Directed a **timing attack** on *Intel 7th gen i7* processor to reverse engineer the confidential DRAM address mappings
- Established illicit inter-process communication through a **covert channel** by utilising the address mappings

Decentralised Professional Networking Application | *Blockchain Technology* Ongoing

- Leveraging the Ethereum blockchain to make a **smart contract** for professional networking in *Solidity*
- Ensuring public verifiability of online identity by integrating **Keybase** and hence removing the need for trust

EYE, The Interpreter that Visualises | *Software Systems Lab* Autumn 2017

- Developed an interpreter in Python for a subset of C language with **visual execution** as a debugging tool
- Displayed the **memory layout** (stack variables and heap data structures) after every statement execution

Other Projects

- **Chess Classic:** Designed a chess AI engine using **mini-max** algorithm with **alpha-beta prunning** in *Racket*
- **Big Data Processing:** Analysed sentiments in a large amount of data using RDD and Dataset in **Spark**
- **Maze Solver:** Formulated the problem as a Markov Decision Process and solved using value iteration
- **Distributed Spanning Tree Protocol:** Implemented a distributed system of network bridges communicating with their neighbours through message passing to agree upon a spanning tree of the extended LAN
- **C-like Compiler:** Built a compiler from scratch to compile basic C programs into MIPS64 instructions
- **Railway Signalling Controller:** Programmed a Spartan **FPGA** board in VHDL to work as signal controller
- **Onion Routing:** Analysed its thread model, implementation, security guarantees and side-channel attacks

Technical Skills

Programming Languages	**Proficient:** C/C++, Scala, Java, Python **Competent:** CUDA, Solidity, Bash, VHDL
Web Development	JavaScript, JSP, HTML5, CSS, Bootstrap, PHP
Softwares	Git, Docker, MATLAB, Cassandra, PostgreSql, JDBC, Wireshark

Positions of Responsibility

Department Academic Mentor Apr 2018 - Apr 2019

- Mentor to 6 sophomores for helping them cope up with the curriculum and solve their general concerns

Teaching Assistant IIT Bombay

- Foundations of Data Structures - MOOC course
- Data Structures and Algorithms (Theory + Lab)
- Design and Analysis of Algorithms
- Computer Programming and Utilisation - Core TA

Department Sports Secretary Apr 2017 - Mar 2018

Extracurriculars

- **Wrestle-AI:** Secured *first* place in wrestle-AI competition for engineering an **autonomous wrestling bot** in arduino with line-following and wall-following capabilities using *IR* and *ultrasonic* sensors 2017
- Received certificate of merit for swimming for **8 hours straight** covering **12 km** in Swimmathon 2017
- Successfully completed **80 hours** under the National Service Scheme (**NSS**) which involved teaching science at LCCWA (NGO) and making presentations about basic banking 2016-17
- Engineered an **app-controlled bot** for **XLR8** competition organised by Robotics Club IITB 2016